

Front-End Developer Interview Preparation Guide

This guide contains a comprehensive list of questions and topics to help you prepare for front-end developer interviews. It is especially useful for those working with JavaScript and React.

1. Core JavaScript & React Questions

- Explain closure in JavaScript.
- What is the difference between var, let, and const in JavaScript?
- How does JavaScript's event loop work?
- What are the different ways to handle asynchronous code in JavaScript?
- Explain React's reconciliation algorithm (virtual DOM).
- How does React differ from Angular and Vue?
- What is the purpose of React.memo?
- What is the difference between componentDidMount and useEffect?
- Explain the useRef hook and its use cases.
- What are the benefits of using functional components over class components?
- How would you manage state in a large React app?

2. Performance & Optimization

- How would you optimize the performance of a React application?
- What is code-splitting in React, and how would you implement it?
- How do you handle lazy loading of components in React?
- What is memoization, and how do you apply it in React?
- What is the difference between shallow rendering and deep rendering?
- How do you ensure the accessibility of your web app?
- What strategies would you use to reduce the load time of a web page?

3. React Router & State Management

- How does React Router work? Can you explain the use of Route, Switch, and Link?
- What is Redux, and when would you choose it over Context API?
- What are the main principles of Redux?
- How do you handle forms in React, and what are controlled vs uncontrolled components?
- What is the role of React Context? When would you use it over Redux?

4. HTML, CSS & Responsive Design

- What is the box model in CSS?
- What are CSS preprocessors like SASS and LESS, and why would you use them?
- How do you handle browser compatibility issues?
- Can you explain CSS Grid and Flexbox with examples?
- What is the difference between position: absolute and position: fixed?

- • Explain the concept of z-index in CSS.

5. Testing & Debugging

- • What are the different testing strategies you follow in React applications?
- • Explain how you would test a React component.
- • What is Jest, and how is it used for testing React components?
- • How do you debug React applications effectively?
- • What is the role of PropTypes in React?

6. System Design & Architecture

- • How would you design a scalable React application?
- • What are the considerations for managing large-scale state in React?
- • How do you handle routing in a multi-page application (MPA)?
- • How do you structure the folders and components in a large React application?

7. Behavioral Questions

- • Can you describe a challenging project you worked on, and how you handled the challenges?
- • How do you stay up-to-date with new technologies in the front-end ecosystem?
- • How do you manage deadlines and multiple tasks in a fast-paced environment?
- • Describe a time when you had to collaborate with designers or back-end developers. How did you handle it?